TNBC Drug Sensitivity & Selectivity Platform

Interdisciplinary Research Blueprint: Pharmacogenomics, Machine Learning & Biochemical Validation

PRD V1.0 ALIGNED

1. EXECUTIVE SUMMARY, SCIENTIFIC PREMISE & SCOPE LOCK

Core Objective: Develop an auditable machine learning model to predict and rank fitted drug sensitivity (Area Under Curve, AUC) for Triple-Negative Breast Cancer (TNBC) from baseline transcriptomic profiles, followed by rigorous in-vitro cytotoxicity validation.

Scope Clarification: The system ranks existing well-annotated compounds from GDSC2 (~297 oncology drugs). It is *not* an open-ended generative tool or clinical therapy decision engine; it is an experimental drug-repurposing and response prioritization pipeline.

The Biological Selectivity Hypothesis: Top computationally prioritized compounds must exhibit preferential cytotoxicity against aggressive TNBC cells (**MDA-MB-231**: ER-, PR-, HER2-) over non-TNBC hormone-responsive control cells (**MCF-7**: ER+, PR+, HER2-), quantified via the therapeutic window ratio: `Selectivity Index (SI) = IC50(MCF-7) / IC50(MDA-MB-231) > 1.5`.

2. HIGH-DIMENSIONALITY STRATEGY & CROSS-CANCER TRANSFER

Data Scarcity & Transfer Learning: Publicly available TNBC cell lines are limited (n ≈ 25–35), rendering models trained solely on TNBC prone to catastrophic overfitting. Training encompasses **Pan-Cancer + Pan-Breast models** (n > 800 cell lines, ~150k drug-response pairs) to learn general cytotoxic response dynamics, while evaluation is strictly isolated on the TNBC slice.

Dimensionality Reduction (p >> n): Whole-transcriptome data (~19,000 genes) is filtered down to the top 2,000 high-variance genes, normalized using `log2(TPM + 1)` and standardized (z-score). PCA is fitted exclusively within training folds (128 components) to avoid data leakage, combined with drug one-hot encoding for GDSC2 agents.

Target Metric Choice: Target is **fitted AUC** (bounded [0, 1], lower indicates higher sensitivity). Natural log IC50 (`LN_IC50`) serves as a secondary reference metric.

3. CROSS-DISCIPLINARY STREAM RESPONSIBILITIES

Bioinformatics Stream

- **ID Harmonization:** Cross-reference DepMap Model IDs, Sanger Passports, and COSMIC IDs.
- **Subtype Annotation:** Curate clinical ER/PR/HER2 status to reliably tag the TNBC cohort.
- **Early Shortlisting:** Deliver an early candidate list by Week 6 to initiate chemical vendor procurement.
- **Biomarker Concordance:** Match SHAP-derived driver transcripts to known cancer pathways.

Applied ML & Software

- **Leakage Prevention:** 5-Fold `GroupKFold` on `cell_line_id` ; MDA-MB-231 & MCF-7 fully held out.
- **Model Benchmarking:** Benchmark DummyMean baseline against RandomForest, XGBoost, and CatBoost.
- **Model Interpretability:** Compute global and cell-specific SHAP values (top 20 feature drivers).
- **Serving & GUI:** Deploy FastAPI endpoints (`/predict` , `/rank`) and Streamlit dashboard.

Biochemistry / Wet-Lab

- **Early Procurement:** Audit lab reagents, media, and order top candidate drugs by Week 6–8.
- **Model Maintenance:** Culture and verify morphology of MDA-MB-231 and MCF-7 cell models.
- **Viability Assays:** Run 72h MTT / CellTiter-Glo assays (6–8 doses, 3 passages, triplicates).
- **Controls & 4PL:** Include Doxorubicin (positive control) and DMSO (vehicle); fit 4PL curves.

20-Week Synchronized Implementation Roadmap

Phased Engineering, Procurement & Wet-Lab Experimental Milestones

EXECUTION TIMELINE

4. PHASED MULTI-DISCIPLINARY WORK BREAKDOWN

Period	Bioinformatics Stream	Applied Machine Learning	Biochemistry & Pharmacology
Weeks 1 – 2	Freeze DepMap (24Q2+) and GDSC2 (v8.x). Harmonize identifiers via COSMIC & Sanger Passports.	Construct automated audit script (`audit.py`); inspect missingness and target AUC distributions.	**Lab Audit:** Inventory MTT kits, DMEM/RPMI media, multi-channel pipettes, and obtain budget approval.
Weeks 3 – 5	Curate TNBC receptor status; filter top 2,000 high-variance genes; verify MDA-MB-231/MCF-7 profiles.	Build transformation pipeline (log2 TPM, z-score); fit train-isolated PCA (128 dims); export `gold.parquet` .	Thaw, passage, and verify growth rates and morphology of MDA-MB-231 and MCF-7 cell lines.
Weeks 6 – 8	**Gate 2A:** Generate preliminary literature & data-informed drug shortlist (~10 candidates).	Train DummyMean, XGBoost, CatBoost using 5-Fold `GroupKFold` ; log runs to MLflow.	**Early Purchasing:** Order Top 3–5 candidate compounds, Doxorubicin (pos control), and DMSO.
Weeks 9 – 10	Correlate top SHAP predictive genes with biological pathways (DNA repair, PI3K/Akt/mTOR, apoptosis).	**Gate 2B:** Freeze champion CatBoost model; export blind rankings for MDA-MB-231 vs MCF-7.	Optimize 96-well seeding densities (cells/well); conduct DMSO vehicle tolerance tests (≤ 0.5% v/v).
Weeks 11 – 12	Review GDSC2 concentration ranges to advise experimental microplate dosing (nM to µM).	Build FastAPI serving endpoints (`POST /predict` , `POST /rank`); pin Colab dependencies.	Receive compounds; prepare stock solutions in DMSO; formulate 6–8 point serial dilution matrices.
Weeks 13 – 16	Track assay run metadata, compound lot numbers, and cell passage indices (P3–P8).	Develop interactive Streamlit application; connect prediction backend and interactive SHAP plots.	**Assay Execution:** Perform 72h MTT cell viability assays across 3 independent passages in triplicates.
Weeks 17 – 18	Map experimental response discrepancies back to genomic features (e.g., TP53 mutational status).	Automate 4-Parameter Logistic (4PL) regression in `curve_fitting.py` to compute experimental IC50 & AUC.	**Gate 3:** Measure microplate absorbance (OD 570 nm); compile normalized viability matrices.
Weeks 19 – 20	Synthesize biological mechanism of action (MoA) concordance for defense presentation.	Integrate experimental curves into UI; plot Predicted vs Experimental correlation (Spearman ρ).	Compute Selectivity Indexes (SI = IC50_MCF7 / IC50_MDA231); co-author final graduation defense report.

Operational Governance, Data Contracts & Benchmarks

Inter-Team Quality Gates, Data Schemas & Risk Mitigation Protocols

GOVERNANCE & QA

5. INTER-DISCIPLINARY DATA HANDOFF GATES

Gate 1 • Week 5
Bioinfo → ML: Delivery of verified `gold_dataset.parquet` (TNBC annotations, cleaned TPM values).

Gate 2A • Week 6
Bioinfo → Lab: Delivery of `procurement_shortlist.csv` for commercial vendor ordering.

Gate 2B • Week 10
ML → Lab: Delivery of final `model_rankings.csv` : Model-ranked drugs, expected AUC, and SHAP genes.

Gate 3 • Week 18
Lab → ML: Delivery of `raw_absorbance.csv` for automated 4PL curve fitting and rank validation.

6. STANDARDIZED DATA SCHEMAS

CELL_MODEL_METADATA

```
cell_line_id: string (ACH-ID / COSMIC)
cell_name: string (MDA-MB-231 / MCF-7)
subtype: string (TNBC / Luminal_A)
receptor_status: string (ER-,PR-,HER2-)
holdout_status: string (Unseen_Test_Fold)
```

FEATURE_MATRIX_ROW

```
cell_line_id: string
pca_features: float32[128]
drug_id: int (GDSC drug identifier)
target_auc: float32 (Fitted GDSC2 AUC)
target_ln_ic50: float32 (Secondary label)
```

MODEL_INFERENCE_OUTPUT

```
cell_line: string
model_version: string (catboost-v1.0)
predicted_auc: float32 (0.0 to 1.0)
sensitivity_rank: int (1 = most sensitive)
top_shap_transcripts: string[]
```

BIOCHEMICAL_ASSAY_READOUT

```
cell_line: string
compound_name: string
dose_series: float32[] (6-8 points, μM)
fitted_ic50: float32 (μM, 4PL Hill fit)
selectivity_index: float32 (IC50_MCF7/MDA)
```

7. BENCHMARKS & ACCEPTANCE CRITERIA

Computational Model Gates:

- **Baseline Superiority:** Must strictly outperform the DummyMean baseline across RMSE, MAE, and Spearman ρ.
- **Prediction Quality:** Test set $R^2 > 0.25$ and per-cell-line Spearman rank correlation $\rho > 0.40$ on the breast/TNBC slice.

Biochemical & Experimental Gates:

- **Dose-Response Separation:** Clear sigmoidal separation between positive control (Doxorubicin) and vehicle (DMSO).
- **Selectivity Confirmation:** `SI > 1.5` for ≥ 1 top candidate (SI > 2.0 indicates high selectivity).
- **Rank Concordance:** Positive Spearman correlation ($\rho > 0$) between Predicted AUC and Experimental AUC.

8. SCIENTIFIC DEFENSE & CONTINGENCY PROTOCOLS

Experimental Risk Mitigation:

If assay variance or cell line drift weakens continuous correlation ($\rho < 0.20$), the defense narrative pivots cleanly to robust scientific anchors:

1. Binary Classification: Group drugs into Sensitive (AUC < 0.5) vs Resistant (AUC ≥ 0.5) to evaluate classification accuracy rather than continuous rank order.

2. Assay Control Validity: Demonstrate rigorous assay execution via expected IC_{50} values for Doxorubicin against published literature ranges.

3. Pathway Concordance: Provide qualitative mechanistic validation aligning SHAP driver transcripts with established compound targets.